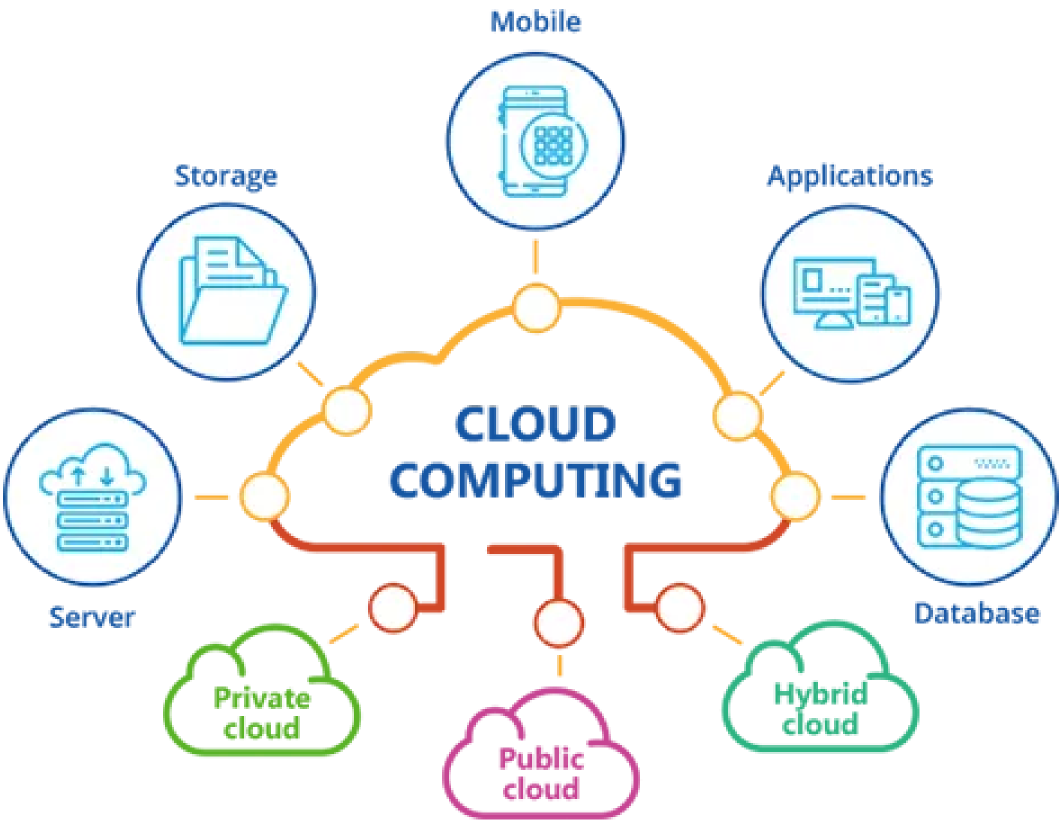
Cloud Computing



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Introduction to Cloud Computing

Cloud Computing provides us a means by which we can access the applications as utilities, over the Internet. It allows us to create, configure, and customize applications online.

What is Cloud?

The term Cloud refers to a Network or Internet. In other words, we can say that Cloud is something, which is present at remote location. Cloud can provide services over network, i.e., on public networks or on private networks, i.e., WAN, LAN, or VPN. Applications such as e-mail, web conferencing, customer relationship management (CRM), all run in cloud.

What is Cloud Computing?

Cloud computing refers to the delivery of computing services—including servers, storage, databases, networking, software, analytics, and more—over the internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale. Instead of owning and maintaining physical hardware or infrastructure, users access these resources on-demand from a cloud service provider.

Key Concepts:

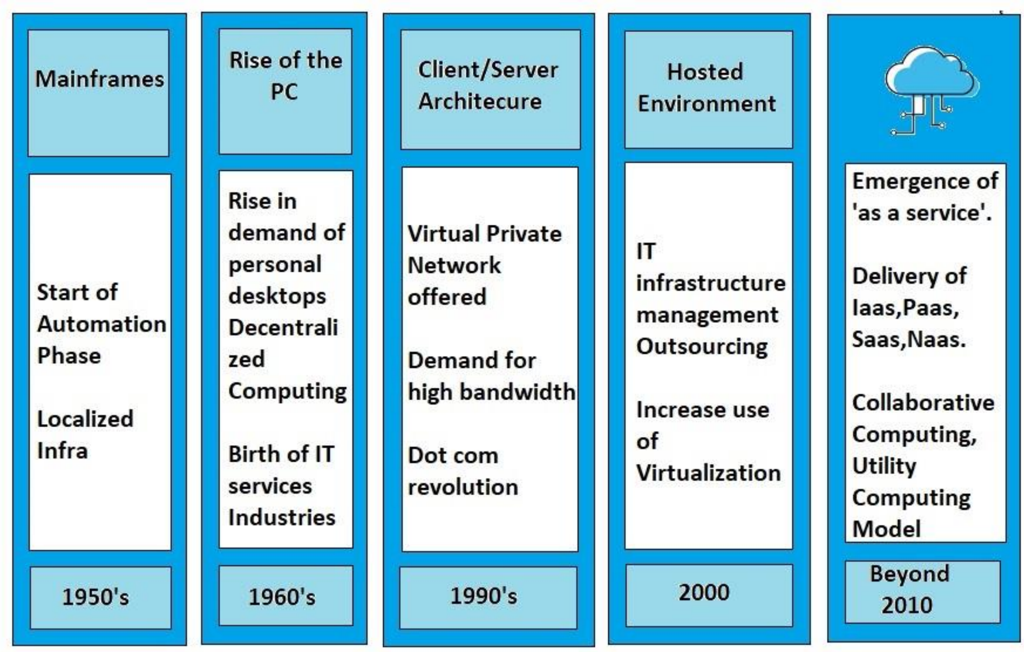
1. On-Demand Service: Resources can be provisioned and accessed instantly, scaling up or down based on demand.

2. Resource Pooling: Multiple users share the same infrastructure, allowing for efficient utilization and cost-sharing.

3. Elasticity: The ability to scale resources automatically to accommodate fluctuating workloads.

4. Measured Service: Cloud systems automatically control and optimize resource use by metering usage, allowing for pay-as-you-go or subscription-based models.

Cloud Computing History:



Early Foundations (1950s - 1990s):

- 1950s-1960s: The concept of utility computing emerges, where computing resources are provided as a service.
- 1970s-1980s: IBM develops virtual machines, allowing multiple operating systems to run simultaneously on a single physical machine.
- 1990s: Telecommunications companies begin offering virtual private network (VPN) services with secure access to shared systems.

Evolution and Emergence (Late 1990s - Early 2000s)

- Late 1990s: Salesforce pioneers the delivery of enterprise applications via a website, a precursor to Software as a Service (SaaS).
- Early 2000s: Amazon Web Services (AWS) launches in 2006, offering cloud-based services like storage and computation on a pay-as-you-go basis.
- 2006: Google releases Google Apps, marking the entry of a major player into cloud-based productivity tools.
- 2008: Microsoft Azure is officially announced by Microsoft's CEO, Steve Ballmer, and its development begins around this time. However, Azure becomes commercially available in February 2010.

Expansion and Adoption (Mid-2000s - Present)

- 2008: OpenStack, an open-source software platform for cloud computing, is launched, fostering interoperability among cloud services.
- 2010s: Cloud computing becomes more mainstream, with major companies moving infrastructure and services to the cloud.
- 2012: Google rebrands its services under the broader umbrella of Google Cloud Platform (GCP) after the initial launch of Google App Engine in April 2008.
- 2015-Present: The rise of hybrid and multi-cloud environments as organizations seek flexibility and redundancy across various cloud platforms.

Recent Trends (2010s - Present)

- Serverless Computing: Services like AWS Lambda and Azure Functions allow developers to run code without provisioning or managing servers.
- Edge Computing: With the growth of IoT, there's an emphasis on processing data closer to its source to reduce latency, leading to edge computing models.
- Artificial Intelligence (AI) and Machine Learning (ML) Integration: Cloud providers offer AI and ML tools, making these technologies more accessible.

Cloud Service Models:

Cloud services that help businesses transform their digital experience while reducing the infrastructural costs in turn.

Following are a few cloud computing examples and uses:

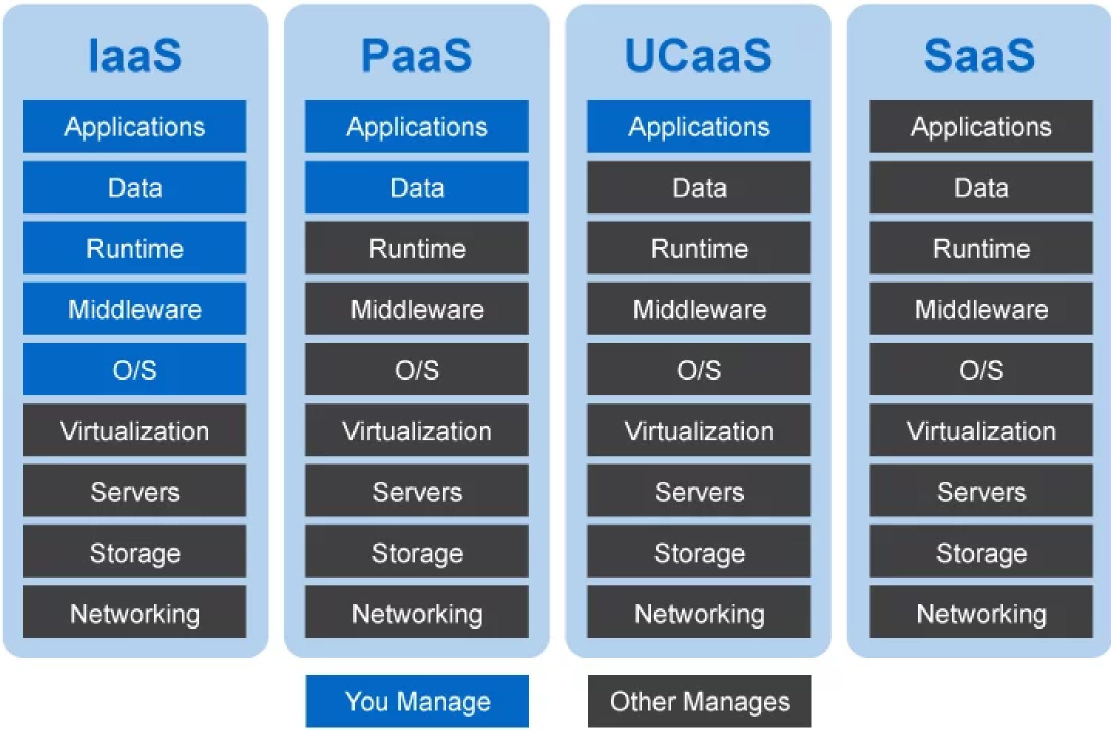
- **Communication:** Skype, WhatsApp, Slack
- **Productivity:** Microsoft Office 365, Gmail
- **Data Storage:** Dropbox, Facebook, Gmail
- **Business Processes:** Salesforce, HubSpot
- **Application Development:** Amazon Lumberyard
- **Big Data Analytics:** Hadoop, Cassandra, HPCC
- **Social Networking:** Myspace, LinkedIn, Twitter

SaaS	PaaS	IaaS	UCaaS
Salesforce	Microsoft Azure	Amazon EC2	Zoom
Microsoft 365	Google App Engine	Microsoft Azure	Microsoft Teams
Pardot Marketing Automation	Windows Azure AppFabric	Google Cloud Platform	Fuze
JIRA	SQL Azure	Magento 1 Enterprise	Google Hangouts
Dropbox	AWS Elastic	Digital Ocean	Jive
Slack	Beanstalk		
Amazon Web Services	Force.com		

Types of cloud services:

IaaS, PaaS, and SaaS all these cloud services differ primarily in what they offer to the end user. Each cloud service has its benefits depending upon the business and functional requirements.

1. **Infrastructure as a Service (IaaS)**
2. **Platform as a Service (PaaS)**
3. **Software as a Service (SaaS)**
4. **Unified Communications-as-a-Service (UCaaS)**



1. Infrastructure as a Service (IaaS): Infrastructure-as-a-service (IaaS) is one of the most pliable cloud computing service models which presents the required infrastructure and all the computing resources to users in an entirely remote environment. The vendors delivering PaaS service are responsible to provide advanced services to address your company's infra needs, which involves data storage space, servers, virtualization, and networking.

IaaS is a highly expandable and lucrative service as outsourcing data centers, cloud components and servers annihilate the necessity for establishing an in-house infrastructure. This service model can aid you to upscale and downscale your infra service on-demand. IaaS will be a good fit for SMEs and startups that might not have ample resources to buy essential infra for building their own network.

2. Platform as a Service (PaaS): Platform-as-a-service (PaaS) is a subset of SaaS, and it is built on virtually the same infrastructure as IaaS. Besides that, in this service model, the vendor assists companies with middleware, database management systems, OS, web servers and development tools. All of these encapsulate a remote environment where users can build, compile, and run their software products without requiring any in-house hardware/software.

Thus, PaaS enables users to emphasize their application resources and effectively manage their data as the rest of the things will be handled efficiently by the vendors. Since the platform is web-accessible, remote development unit can have ready access to all the assets ubiquitously. This, in turn, will help them to improve their product development life cycle, resulting in more agile product delivery.

3. Software as a Service (SaaS): Software-as-a-service (SaaS) presents a comprehensive product that is run and administered by a cloud service provider/vendor. SaaS is a service in the cloud, extending an entire software suite in pay-per-use form. It is made accessible to the end-consumers over a ubiquitous network, i.e. the internet.

SaaS is the superset of both, PaaS, and IaaS as it offers the entire package from infrastructure, middleware, OS to applications deployed over the web, that can be seamlessly accessed, invariant to time, place and platform. This fully developed cloud service model encourages businesses and respond expeditiously to it as well as scale business operations remotely.

4. Unified Communications-as-a-Service (UCaaS): IT is trending up amid the current crisis as this service model presents communications continuity and remote collaboration services to users, worldwide, via the cloud network. Moreover, this service model provides advanced security and reliability, enabling the remote workforce to work seamlessly in a secure, virtualized cloud environment.

As work-from-home is becoming the new norm and the digital workplace becoming the future of work, both SMEs and large enterprises are embracing UCaaS services to keep their teams connected while they work apart. UCaaS platform not only empowers the mobile workforce to connect and collaborate over phone/video calls but also enables them to share files, documents, or resources via the cloud infrastructure for streamlined workflows.

Deployment Models:

Cloud Deployment Model functions as a virtual computing environment with a deployment architecture that varies depending on the amount of data you want to store and who has access to the infrastructure.

Types of Cloud Computing Deployment Models:

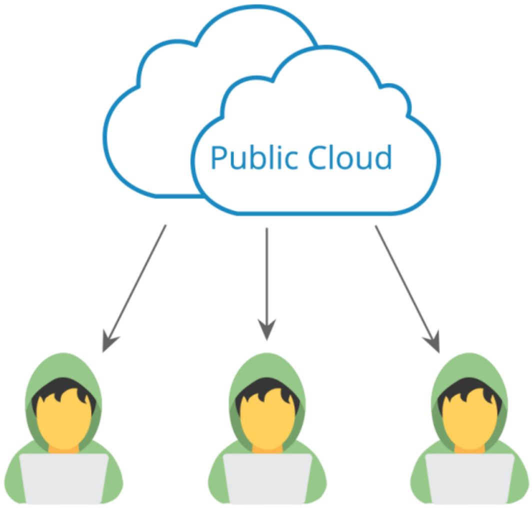
The cloud deployment model identifies the specific type of cloud environment based on ownership, scale, and access, as well as the cloud’s nature and purpose. The location of the servers you’re utilizing and who controls them are defined by a cloud deployment model. It specifies how your cloud infrastructure will look, what you can change, and whether you will be given services or will have to create everything yourself. Relationships between the infrastructure and your users are also defined by cloud deployment types.

1. **Public Cloud**
2. **Private Cloud**
3. **Community Cloud**
4. **Hybrid Cloud**

1. Public Cloud: A public cloud is the most common type of cloud. It's also where most people's minds go to when they think of cloud technology. In a public cloud, your data is stored across remote servers that are hooked together to function as one, all-encompassing network. These servers are not owned by you but rather by a third-party provider.

When you use a public cloud, think of it as renting space on someone else's server. The "who" that you are renting from are cloud service providers. Some major cloud service providers are Amazon, Google, and Microsoft. These companies own and manage all the hardware, software, and infrastructure required for running the cloud.

The key differentiator with a public cloud is that anyone can use it. All you have to do is sign up with a cloud service provider, and you will have access to the cloud. This means that your data uses the same resources as every other customer of that cloud provider.



Advantages:

• **Low Cost** – The service providers are the ones who fund the entire infrastructure (no need for you to invest in hardware or software). And aside from the initial fee, you're good to go on the 'pay per use' policy under this model, which means, no overhead costs.
• **Convenient Infrastructure Management** – You don't need to develop or maintain your software because the service provider does that for you. The setup and use are convenient.
• **Time** – It reduces time in developing, testing, and launching new products. Aside from that, the extensive network of your provider's servers ensures that your infrastructure is available for use 24X7.

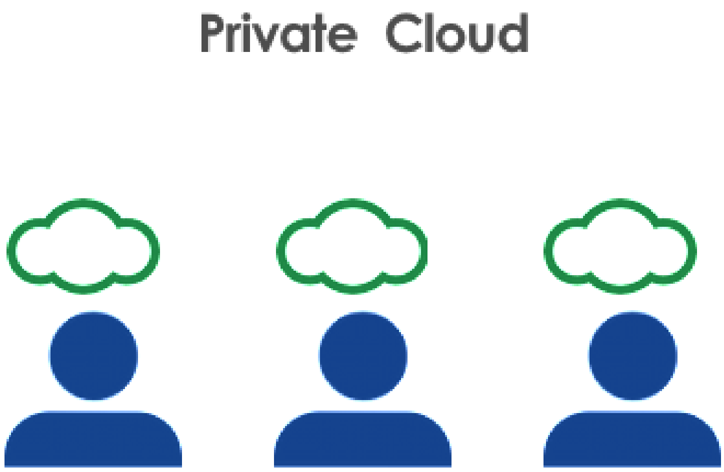
Disadvantages:

• **Data and Security Risk** – As the name suggests, it's a public platform for insensitive data storage. As a result of shared resources, security risks are high due to their vulnerability. A user is deprived of knowing where their information is being stored and who all have the access to it.
• **Network Performance** – Due to an increase in the number of users, the network performance suffers from instability.
• **Lack of Customization** – Since it's a public cloud, personalization will always be limited.

2. Private Cloud: A private cloud, on the other hand, is reserved for a single organization or business. There aren't any shared resources with a private cloud. All hardware and software is dedicated to the owner of the cloud.

You can create a private cloud using owned resources, such as a data center, or you can also use a third-party provider. If you choose a third-party provider, specific hardware and software are dedicated solely to your organization.

Some organizations choose a private cloud over a public cloud for improved flexibility and security. With a private cloud, a business can customize the cloud to meet its specific needs. It also allows for more control when it comes to security because resources are not shared.



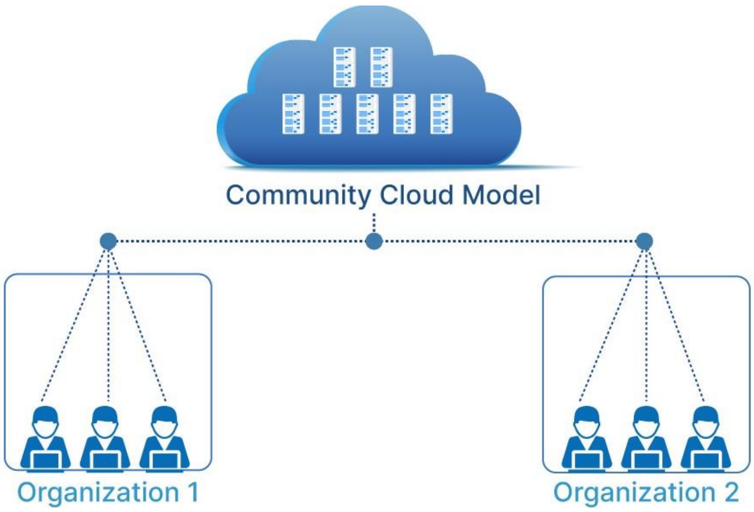
Advantages

- **Customization** – Unlike the public cloud deployment model, a company using a private cloud can customize its solution as per its own requirements.
- **Data and Security** – The security of data is high, as in, only authorized internal personnel is allowed to access data. This makes it ideal for storing corporate data. A company can also separate the sets of resources on the same infrastructure which makes it even more secure.
- **Control** – Your organization is the exclusive owner. You get absolute control over service integrations, IT operations, rules, and user practices.
- **Legacy Systems** – A special feature about private clouds is that it supports legacy applications that are not functional on public clouds.

Disadvantages

- **High Cost** – Yes! You'll need to invest in hardware and software. Aside from that, there are also expenses for in-house staff and training. And not to forget maintenance, which is quite high.
- **Limited Scalability** – Private clouds are scaled within internal limited hosted resources. Scalability also depends on the choice of your underlying hardware.

3. Community Cloud: A community cloud deployment model resembles a private cloud, the only difference being the set of users. While a private cloud implies that one company owns the server, in the case of a community cloud, it is several organizations with similar backgrounds. Companies share the infrastructure and related resources on a community cloud. This model supports joint business organizations, ventures, research organizations, and tenders, etc. Community Cloud is a way to preserve the benefits of economy of scales with the private cloud.



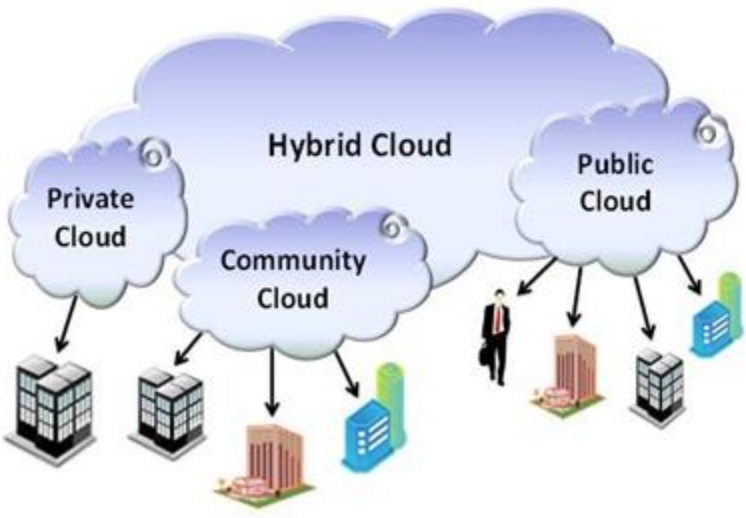
Advantages

• **Cost** – A community cloud is a cheaper way to avail the benefits of a private cloud. That's because multiple companies can share the bill together.
• **Data Sharing and Collaboration** – With a community cloud, sharing data becomes easy and its collaborative space allows clients to enhance their efficiency. Aside from that, configuration and protocols within a community system meet the needs of the specific industry.

Disadvantages

• **Limited Storage and Bandwidth**– Within community systems, limited storage and bandwidth capacity become a common problem.

4. Hybrid Cloud: A hybrid cloud is a combination of public clouds, private clouds, and on-premises infrastructure. A business using a hybrid cloud model often uses a public cloud for basic computing tasks, while storing more sensitive data on a private cloud or an on-site server. For many companies, a hybrid cloud model is the best of both worlds.



Advantages

• **Flexibility and Control** – Companies can choose to allocate resources in accordance with specific cases.
• **Cost** – Since public clouds provide scalability, you're only liable to pay for the extra capacity and that too in case you need it.
• **Agility** – In times like today, agility is what brings productivity and progress. With hybrid cloud deployment, your organization will have the opportunity to develop and test new applications in the right period.

Disadvantages

• **Maintenance**– Agree or don't, a hybrid cloud computing model requires more maintenance which means, a higher operating expense for your organization.
• **Challenging Integration** – Data and application integration be quite challenging when you're building a hybrid cloud. It's also true that activating two or more infrastructures will cover a high initial cost.

Key Service Providers:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)
- IBM Cloud
- Oracle Cloud

Conclusion:

Cloud computing has revolutionized the way businesses and individuals utilize technology. Its flexibility, scalability, and cost-efficiency have made it a cornerstone of modern IT infrastructure, enabling innovation and agility across industries.